### 2.2.3 Irrational Numbers

**Theorem 2.1:** $\sqrt{2}$ is not a rational number.

*Proof.* Suppose that $\sqrt{2}$ is a rational number. Let $\sqrt{2} = \frac{m}{n}$, where $m$ and $n$ are positive integers with no common factors greater than 1. Then, we have $m^2 = 2n^2$, which implies that $m^2$ is even and hence $m$ is even.

Let $m = 2k$. Then, we have $2n^2 = 4k^2$ which gives $n^2 = 2k^2$.
Thus, $n$ is also even.
It follows, that $m$ and $n$ are even numbers having a common factor 2.
Thus, we arrived at a contradiction.
Hence, $\sqrt{2}$ is an irrational number. ■

**Remark:**

(i) Note that in the above proof we have assumed the contrary of what we wanted to prove and arrived at a contradiction. This method of proof is called '*proof by contradiction*'.
(ii) There are points on the number line that are not represented by rational numbers.
(iii) We call those numbers on the number line that do not correspond to rational numbers as *irrational numbers*. The set of all irrational numbers is denoted by $\mathbb{Q}'$ (For number line see Figure 1.2.)

Every real number is either a rational number or an irrational number, but not both. Thus, $\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}'$ and $\mathbb{Q} \cap \mathbb{Q}' = \emptyset$.

As we already knew that every terminating or infinite periodic decimal is a rational number, we see that the decimal representation of an irrational number will neither be terminating nor infinite periodic. The set $\mathbb{R}$ of real numbers can be visualized as the set of points on the number line such that if $x < y$, then $x$ lies to left of $y$.

Figure 2.2 demonstrates how the square roots of 2 and 3 can be identified on a number line.

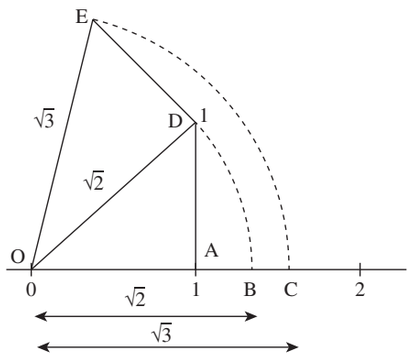



**Figure 2.2**

We notice that $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$.

As we have already observed, irrational numbers occur in real life situations. Over 2000 years ago people in the Orient and Egypt observed that the ratio of the circumference to the diameter is the same for any circle. This constant was proved to be an irrational number by Johann Heinrich Lambert in 1767. The value of $\pi$ rounded off to nine decimal places is equal to $3.141592654$. The values $\frac{22}{7}$ and $3.14$, used in calculations (such as area of a circle or volume of a sphere) are only approximate values for $\pi$.

The number $\pi$, which is the ratio of the circumference of a circle to its diameter, is an irrational number.